Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Data Mining, Dimensional Data Modeling, Star/Snowflake Schemas, Slowly Changing Dimensions (SCD Type 1/2), Semantic Modeling, Self-Service BI, Predictive Modeling, Machine Learning, Data Quality Assurance, Automated Data Auditing, Performance Tuning & Query Optimization, Go-To-Market Strategy Analytics, Quantitative Analysis, Cross-Functional Project Management

Programming Languages & Tools: SQL, Advanced SQL, Python, PySpark, Apache Spark, Azure Data Factory (ADF), Azure Synapse, Azure SQL, Databricks, AWS (S3, Redshift), Snowflake, Power BI (DAX, Measures), Tableau, MicroStrategy (MSTR), Scikit-Learn, XGBoost, SARIMA, Airflow, KIRO IDE

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Data Analyst & Product Analytics Lead — **May 2022 – Present**

- **Architected dimensional data models and semantic layers** utilizing advanced SQL stored procedures, Star/Snowflake schemas, and Slowly Changing Dimensions (SCD Type 1/2) to support scalable product data analytics and revenue optimization initiatives across global markets.
- **Developed executive-ready Power BI and Tableau dashboards** incorporating complex DAX measures and Key Operating Metric (KOM) frameworks to monitor portfolio health, track strategic product initiatives, and deliver actionable insights to cross-functional leadership.
- **Implemented automated data quality audit frameworks and validation firewalls** reducing operational turnaround by over 80% and ensuring stringent data integrity across large-scale databases, cloud pipelines, and downstream analytical assets.
- **Executed predictive customer behavior and forecasting models** using XGBoost, Random Forests, Scikit-Learn, and SARIMA within PySpark and Azure environments, improving operational reliability by 20% and optimizing resource capacity planning.
- **Streamlined automated testing and data pipeline profiling** leveraging Agentic AI and prompt engineering frameworks in KIRO IDE to accelerate reporting feature delivery by 40% while maintaining 99.9% data availability SLAs.

Master of Science in Information Systems & Graduate Certificate in Data Science — **May 2022 – Present**

- **Completed rigorous graduate coursework in advanced analytics** data mining, database management, and quantitative problem-solving with a 3.95/4.0 GPA from the University of Cincinnati.
- **Applied statistical modeling and machine learning techniques** to large-scale datasets, translating complex business challenges into structured quantitative solutions.
- **Built foundational expertise in enterprise data architectures** semantic modeling, and data-driven decision-making methodologies.

Bachelor of Engineering in Petrochemical — **May 2022 – Present**

- **Mastered complex quantitative and mathematical analysis** building a robust technical aptitude for large-scale systems engineering and structured problem-solving.
- **Executed multidisciplinary engineering projects** requiring meticulous attention to operational detail, process optimization, and data-backed evaluation.
- **Developed strong analytical foundations** enabling rapid adoption of modern data stacks, cloud platforms, and data warehousing technologies.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.